



* 1. What is Kubernetes (K8s) ?

- **Definition:** Open-source container orchestration platform for automated scaling and management.
- **Self-Healing:** Restarts failed containers, reschedules nodes, and replaces unresponsive pods.
- **Declarative Config:** You define the desired state; K8s reconciliation loop achieves it.

* Control Plane Components:

kube-apiserver - Hub of all communications **etcd** - Consistent key-value store

kube-scheduler - Assigns pods to nodes **controller-manager** - Regulates cluster state

CONTROL PLANE

- API Server
- etcd DB
- Schedulers

Kubelet

WORKER NODE 1

- Pods (Nginx, API)

WORKER NODE 2

- Kube-Proxy + CNI

* Essential Cluster Health Commands:

```
kubectl cluster-info
kubectl get nodes -o wide
kubectl get pods -A
```



* 2. Pods, Deployments & ReplicaSets :

- **Pod:** Smallest deployable unit in K8s; encapsulates one or more co-located containers.
- **ReplicaSet:** Ensures a specified number of identical pod replicas are running at all times.
- **Deployment:** Provides declarative updates for Pods; enables rolling updates and instant rollbacks.

* Deployment YAML Manifest (deployment.yaml):

```
apiVersion: apps/v1
kind: Deployment
metadata: { name: web-api }
spec:
  replicas: 3
  selector: { matchLabels: { app: web } }
  template:
    metadata: { labels: { app: web } }
```



* 3. Exposing Pods with Services :

- **Problem:** Pod IPs are ephemeral and change whenever a pod restarts.
- **Service:** Stable virtual IP and DNS name that load-balances traffic across matching pods.
- **Service Types:** ClusterIP (internal), NodePort (static port on node), LoadBalancer (cloud external).

Service Type	Accessibility	Use Case
ClusterIP	Inside cluster only	Inter-service / Data
NodePort	NodeIP:NodePort (30000+)	Dev / Direct Node Te
LoadBalancer	External Cloud IP	Production public en
Ingress	HTTP/HTTPS Host routing	Domain SSL & Path pr



* 4. Must-Know kubectl Commands :

Command	Action	Flags
<code>kubectl apply -f .</code>	Deploy all manifests	<code>--dry-run=client</code>
<code>kubectl get pods</code>	List pods in namespace	<code>-n prod -w</code>
<code>kubectl describe p</code>	Inspect events and failures	<code><pod-name></code>
<code>kubectl logs -f</code>	Stream container stdout	<code>--tail 100</code>
<code>kubectl exec -it</code>	Open interactive shell	<code>-- sh</code>
<code>kubectl rollout un</code>	Instant rollback deployment	<code>deployment/web</code>

* Pro Troubleshooting Shortcut:

Port-forward pod to localhost without exposing public

```
kubectl port-forward pod/web-api-7b8f9 8080:80
```

Sort pods by memory usage

```
kubectl top pods --sort-by=memory
```